



# HOW TO BE A **ROBOTICIST**

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A ROBOCLAW Educational Mini-Book

For students exploring industrial robotics careers

**INDUSTRIAL ROBOTICS • CAREERS • SKILLS**

Built around the ROBOCLAW Lab experience

# Welcome to the Path

This book is for every student who has just controlled a real industrial robot in a ROBOCLAW Lab session — or is about to. The machines you move in the lab are the same class of robots that build cars, assemble electronics, package food, and work in laboratories around the world.

## What you will find inside

- The mindset that separates good roboticists from average ones
- Core knowledge every industrial roboticist must master
- Practical skills you can start building today
- Real career paths and what the work actually looks like
- How the ROBOCLAW Lab experience connects to the profession
- *"The best roboticists combine technical skill with curiosity and responsibility. Treating real machines with respect is the first step toward becoming a professional."*

— ROBOCLAW Educational Series

## What Is a Roboticist?

A roboticist designs, builds, programs, tests, and improves robots. In the industrial world, most roboticists work with the kind of machines you control in ROBOCLAW Lab — strong, precise, programmable arms used in factories, laboratories, and production lines.

### Industrial Focus

Industrial roboticists specialize in arms and systems that perform repetitive, precise, or hazardous work: welding, assembly, packaging, machine tending, inspection, and material handling.

### Why It Matters

These robots increase quality, safety, and productivity. They free human workers from dull, dirty, and dangerous tasks so people can focus on higher-skill work.

## The ROBOCLAW Connection

When you move a KUKA KR 10 R1100-2 or an ABB IRB 120 in the lab, you are using the same class of technology that professional roboticists program and integrate every day. The concepts of axes, reach, payload, and repeatability are not classroom theory — they are the daily language of the profession.

# The Roboticist Mindset

## 1. Think in Systems

A robot is never just the arm. It includes the controller, the end effector, safety systems, the part being handled, and the surrounding process. Good roboticists see the whole cell.

## 2. Respect Precision

Industrial robots succeed because they do the same thing the same way every time. Understanding repeatability and accuracy is fundamental.

## 3. Safety First

Real robots can move fast and carry significant force. Designing for safe human-robot interaction is as important as writing good code.

## 4. Iterate Relentlessly

Almost every robot program is improved many times before it is production-ready. Testing, observing, and refining is the real work.

## Curiosity + Responsibility

The strongest roboticists stay curious. They ask why a motion looks the way it does, why a cycle time is limited, and how a small change in tool orientation can improve the process. At the same time they carry responsibility: every program they write affects quality, safety, and the people working near the cell.

### A practical habit

After every lab session or programming practice, write down three things: what worked, what surprised you, and one question you still have. This simple habit builds the observation and documentation skills that professional roboticists use every day.

#### Remember

The robots in ROBOCLAW Lab are real industrial machines.

Treating them with attention and respect is not optional —  
it is the foundation of the profession.

# Core Knowledge: Axes

A typical industrial robot has six rotating joints, called axes. Axes 1–3 move the arm through space (position). Axes 4–6 orient the tool (wrist). Together they allow the tool to reach almost any position and orientation inside the work envelope.

| AXIS | COMMON NAME    | WHAT IT TYPICALLY DOES           |
|------|----------------|----------------------------------|
| A1   | Base rotation  | Swings the entire arm left/right |
| A2   | Shoulder       | Lifts and lowers the upper arm   |
| A3   | Elbow          | Extends and retracts the forearm |
| A4   | Wrist rotation | Twists the wrist                 |
| A5   | Wrist bend     | Pitches the tool up/down         |
| A6   | Tool rotation  | Rolls the end effector           |

## Reach, Payload & Repeatability

### REACH

How far the robot can extend its tool. Measured from the center of the base to the farthest point the tool can reach. The KUKA KR 10 R1100-2 reaches 1101 mm; the ABB IRB 120 reaches 580 mm.

### PAYLOAD

The maximum weight the robot can safely carry at the end of the arm, including the tool itself. Exceeding payload reduces performance and can damage the robot.

### REPEATABILITY

How accurately the robot returns to the same taught position over and over.  $\pm 0.02$  mm means the robot can return to a point within the thickness of a human hair.

## Coordinate Systems & Mounting

Roboticists constantly work with different coordinate frames: the robot base, the tool (end effector), and the world or fixture. Understanding how these frames relate is essential for accurate programming.

### Mounting Flexibility

Both the KUKA KR 10 R1100-2 and the ABB IRB 120 can be mounted on the floor, wall, or ceiling (and often at any angle). Mounting choice changes the available workspace and can solve space or process constraints. Learning to visualize the work envelope in different orientations is a valuable skill.

#### Key takeaway

Reach and payload are not independent. A robot can usually carry its maximum payload only near the center of its workspace.

Always check the manufacturer load diagram.



## Skills You Can Build Now

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- **Observation**

Watch how the robot moves. Notice acceleration, deceleration, and how the wrist orients. Good roboticists see details that others miss.

- **Clear Communication**

Describe motion and problems precisely. 'It looks wrong' is not useful. 'Axis 5 is approaching its limit near the pick point' is.

- **Problem Isolation**

When something does not work, change one variable at a time. This is the scientific method applied to robots.

- **Documentation**

Record what you tried and what worked. Future you (and your teammates) will thank you.

- **Continuous Learning**

Technology evolves. Controllers, software, and safety standards change. The best roboticists keep learning.

## Programming Mindset

Whether you eventually program in KUKA KRL, ABB RAPID, or another language, the underlying thinking is similar: define points, define paths, control speed and acceleration, handle signals, and build robust error recovery.

### From Lab to Profession

In ROBOCLAW Lab you experience the physical reality of moving a real arm. That experience is rare and valuable. Most students only see robots in videos or simulations. You are already ahead because you have felt the difference between a commanded move and the actual motion of a machine.

#### Practice tip

After each lab, try to explain to someone else exactly what the robot did and why. Teaching is one of the fastest ways to deepen your own understanding.

## A Day in the Life

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Industrial roboticists work in many environments: system integrators, automotive plants, electronics factories, food packaging lines, research labs, and equipment manufacturers. A typical day can include:

- Reviewing a new application requirement with a customer or process engineer
- Simulating a robot path in software before touching the real machine
- Teaching points on the shop floor with a teach pendant
- Tuning speeds, accelerations, and zone data for cycle time and quality
- Integrating vision, force sensing, or conveyor tracking
- Writing and testing safety logic and recovery routines
- Documenting the final program and training operators

*The mix of desk work, simulation, and hands-on time is one of the most attractive aspects of the career for many people.*

# Career Paths

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## Robot Programmer / Integrator

Writes the actual motion programs, integrates the robot with fixtures, vision, and PLCs. Often works on-site during commissioning.

## Application Engineer

Helps customers choose the right robot and design the process. Bridges sales, engineering, and the shop floor.

## Automation Specialist

Broader role that may include robots, conveyors, sensors, and complete cells. Common in larger manufacturing companies.

## Maintenance & Reliability

Keeps robots running. Diagnoses mechanical, electrical, and software issues. Critical for uptime.

## Simulation / Digital Twin

Creates virtual models of cells to test programs and layouts before hardware exists. Growing field.

## System Designer

Architects the overall solution — robot choice, tooling, safety, cycle time, cost.

## How People Get Started

Many industrial roboticists begin with a technical high-school program, community-college automation certificate, or engineering degree. Others start as electricians, machinists, or maintenance technicians and move into robotics through on-the-job training.

### What employers look for

- Hands-on experience (even lab or classroom) with real industrial robots
- Understanding of basic mechanics, electricity, and programming concepts
- Safety awareness and a careful approach to equipment
- Willingness to learn specific brands (KUKA, ABB, Fanuc, Yaskawa, etc.)
- **Your advantage** Clear communication and documentation habits

Students who complete ROBOCLAW Lab sessions already have something many entry-level candidates lack: real time controlling industrial arms. Keep a record of your sessions.

## The ROBOCLAW Lab Advantage

Most people who enter industrial robotics first encounter robots only in videos or simplified simulations. ROBOCLAW Lab gives you something different: direct control of real industrial arms (KUKA and ABB) with the same axes, reach, and precision used in factories.

**Physical intuition** You feel the difference between a smooth path and a jerky one.

**Vocabulary that sticks** Axes, payload, repeatability stop being abstract words.

**Respect for the machine** You learn that real robots demand attention and care.  
**Suggestion**

**Portfolio material** Documented lab experience is concrete evidence of interest and skill.  
After each lab session, write a short note: which robot you used, what motion you practiced, and one thing you learned. Over time this becomes a personal learning log that shows growth.

## The Two Lab Robots

### KUKA KR 10 R1100-2

Reach: 1101 mm | Max payload: ~11 kg | Repeatability:  $\pm 0.02$  mm

6 axes | Weight  $\approx$  55–56 kg | Floor / wall / ceiling mounting

Protection: IP65 / IP67 | High-speed packaging, assembly, machine tending

This is a compact high-performance arm from KUKA's Agilus family —  
the same class of robot used in electronics and light manufacturing.

### ABB IRB 120

Reach: 580 mm | Payload: 3 kg (up to 4 kg vertical wrist)

6 axes | Robot mass: only 25 kg | Extremely compact footprint

Common in laboratories, electronics, and tight production cells

One of the smallest multipurpose industrial robots when introduced.

Excellent example of how far small, precise arms have advanced.

## Safety First — Always

Industrial robots can move faster and with more force than most people expect. Professional roboticists treat safety as a non-negotiable foundation, not an afterthought. The same mindset applies in the lab and on the factory floor.

- Never assume a robot is powered off. Verify.
- Understand the work envelope before approaching.
- Respect reduced-speed and collaborative modes when present.
- Keep clear of pinch points and the path of the arm.
- Follow all lab and facility safety procedures without exception.

- **Professional reality.** If something feels unsafe, stop and ask.

Safety standards (ISO 10218, ISO/TS 15066, and local regulations) shape every industrial installation. Learning to think about risk is part of becoming a roboticist.



# Inspiration from the Field

## HISTORICAL CONTEXT

The first industrial robot, Unimate, started work in 1961 removing hot die-castings that were dangerous for people. Every robot since has continued that mission: take the risk and the repetition so humans can do higher-value work.

## PRECISION

Modern industrial robots can return to a position with accuracy better than the thickness of a human hair — and do it thousands of times without fatigue.

## LONGEVITY OF FUNDAMENTALS

The classic 6-axis articulated design introduced in the 1970s is still the workhorse of factories worldwide. Understanding it deeply remains one of the best investments a student can make.

## A Brief Timeline

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|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| <b>1961</b>   | Unimate begins work at GM — first commercial industrial robot     |
| <b>1969</b>   | Stanford Arm — first electrically powered computer-controlled arm |
| <b>1973</b>   | KUKA FAMULUS — first six-axis electromechanically driven robot    |
| <b>1974</b>   | ASEA IRB 6 — first fully electric microprocessor-controlled robot |
| <b>1978+</b>  | PUMA and similar arms become standards for precision assembly     |
| <b>1990s</b>  | Parallel (Delta) robots and better controllers mature             |
| <b>2009</b>   | ABB IRB 120 introduced — one of the smallest multipurpose arms    |
| <b>2010s+</b> | Agilus-class compact high-speed robots (including KR 10 R1100-2)  |

*Full history with citations is available in the ROBOCLAW educational package.*

## Next Steps You Can Take

- 1. Keep a lab log** After every ROBOCLAW session write what you did and what you noticed.
- 2. Master the vocabulary** Axes, reach, payload, repeatability, end effector, controller — use the words correctly.
- 3. Study the two robots** Read the KUKA and ABB fact sheets in the package. Compare them.
- 4. Play the educational games** They reinforce the same concepts used by professionals.
- 5. Explore manufacturer sites** KUKA and ABB publish excellent educational and application material.
- 6. Look for local opportunities** Community colleges, makerspaces, and company tours can deepen exposure.
- 7. Stay curious and careful** The combination is the mark of a future professional.

## Resources Inside This Package

- Pre-lab and post-lab reading packets
- Key vocabulary list
- Matching quiz and printable short test (black-and-white friendly)
- Crossword and word-search puzzles
- Extended factoid list
- Educational games (printable full-page versions)
- Mini info books on key topics
- Full history of industrial robotics with citations
- Public links to official KUKA and ABB specification PDFs
- **All materials are designed to sit around the live lab**
- This mini-book: How to Be a Roboticist

They require no special software and are ready for homeschool, classroom, or independent use. Print the black-and-white versions to save ink when needed.

## You Are Already on the Path

Most people who eventually become industrial roboticists look back and realize the turning point was a moment when a real robot became real to them — when they stopped seeing it as a video and started seeing it as a machine they could understand and control.

If you have completed even one ROBOCLAW lesson, a turning point has already happened for you. The rest is practice, study, curiosity, and the decision to keep going.

The factories, laboratories, and production lines  
of the future will be programmed by people who  
started exactly where you are now.

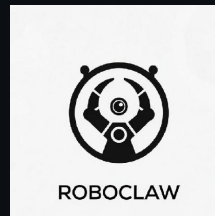
**Keep learning. Keep building. Keep going.**

## Thank You for Exploring

Industrial robotics is a career that rewards precision, curiosity, and responsibility in equal measure.

### ROBOCLAW Educational Series

Built to support students, parents, and teachers around the live ROBOCLAW Lab experience.



All technical specifications drawn from manufacturer data.  
Historical information from IEEE Spectrum, IFR, and company records.

## Visualizing the Six Axes

When you sit at a ROBOCLAW Lab station and move the robot, try to identify which physical joint is moving for each axis command. This mental mapping is one of the earliest skills professional programmers develop.

- A1 movement swings the whole arm — easy to see from above.
- A2 and A3 change the height and extension of the arm.
- A4–A6 are smaller and closer to the tool — watch the wrist carefully.
- **Lab exercise idea** Many singularities and reach limits become obvious only when you move the real machine.

Move only one axis at a time and name it out loud. Then try to reach a point using the minimum number of axes. These simple drills build the spatial intuition professionals rely on.

## Why This Career Lasts

Industrial robotics has been growing for more than sixty years and continues to expand into new industries. The fundamental skills — understanding kinematics, safety, process, and programming — transfer across brands and applications.

Even as collaborative robots, mobile robots, and AI-assisted programming appear, the need for people who deeply understand how a six-axis arm actually moves and interacts with the physical world remains strong.

### Long-term view

Students who invest in real industrial robot experience today are building a foundation that will still be relevant a decade from now. The machines change; the need for skilled people who understand them does not.



# The Future Is Built by People

who understand machines.

**You can be one of them.**



ROBOCLAW Educational Series  
How to Be a Roboticist • 25-page edition